Simple linear regression

In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

In [3]:
```
dataset = pd.read_csv(r"C:\Users\sande\OneDrive\Desktop\Naresh IT\data science and
dataset
```

Out[3]:

	YearsExperience	Salary
0	1.1	39343
1	1.3	46205
2	1.5	37731
3	2.0	43525
4	2.2	39891
5	2.9	56642
6	3.0	60150
7	3.2	54445
8	3.2	64445
9	3.7	57189
10	3.9	63218
11	4.0	55794
12	4.0	56957
13	4.1	57081
14	4.5	61111
15	4.9	67938
16	5.1	66029
17	5.3	83088
18	5.9	81363
19	6.0	93940
20	6.8	91738
21	7.1	98273
22	7.9	101302
23	8.2	113812
24	8.7	109431
25	9.0	105582
26	9.5	116969
27	9.6	112635
28	10.3	122391
29	10.5	121872

In [4]:
```python
x = dataset.iloc[:, :-1]
y = dataset.iloc[:, -1]
```

In [5]:
```python
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2,random_stat
```

In [6]:
```python
from sklearn.linear_model import LinearRegression
regressor = LinearRegression()
regressor.fit(x_train, y_train)
```

Out[6]:

▾ LinearRegression

▼ Parameters

	fit_intercept	True
	copy_X	True
	tol	1e-06
	n_jobs	None
	positive	False

In [7]:
```python
y_pred = regressor.predict(x_test)
```

In [8]:
```python
plt.scatter(x_test, y_test, color = 'red')  # Real salary data (testing)
plt.plot(x_train, regressor.predict(x_train), color = 'blue')  # Regression line fr
plt.title('Salary vs Experience (Test set)')
plt.xlabel('Years of Experience')
plt.ylabel('Salary')
plt.show()
```


```
In [9]: m = regressor.coef_
        c = regressor.intercept_
```

```
In [10]: (m*12) + c
         (m*20) + c
```

```
Out[10]: array([213031.60168521])
```